✈ DGCA CPL / ATPL STUDY NOTES

CHAPTER 3

Modulation

Radio Navigation & Aids — Ground Training Series

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1. Introduction

> **What this section covers:** The definition of modulation, its purpose, and a summary of the five forms of modulation used in aviation: Keyed, AM, FM, Phase, and Pulse.

Modulation is the process of adding information to a radio wave, or the formatting of radio waves for other purposes. It is the critical step that makes radio useful for communications and navigation.

Before audio information can be transmitted, sound must first be converted to an electrical signal by a **microphone** (a device that converts sound waves to electrical current). This electrical signal is the **audio frequency (AF)**; the radio wave it is combined with is the **carrier wave** at **radio frequency (RF)**.

> 🔵 **Five Forms of Modulation Used in Aviation**
>
> 1. **Keyed Modulation** — interrupting the carrier (Morse code; NDB ident)
> 2. **Amplitude Modulation (AM)** — AF amplitude varies RF amplitude
> 3. **Frequency Modulation (FM)** — AF amplitude varies RF frequency
> 4. **Phase Modulation** — AF varies phase of carrier (used in GPS, MLS)
> 5. **Pulse Modulation** — intermittent carrier bursts (radar, DME)

Modulation is not only used for voice communications. Navigation systems such as VOR, ILS, DME, and GPS use modulation for bearing determination, distance measurement, and data transmission.

2. Keyed Modulation

> **What this section covers:** The simplest form of modulation — interrupting the carrier wave to create Morse code symbols.

The simplest way to convey information on a carrier wave is to **interrupt** it — creating short and long bursts of RF energy. This is **keyed modulation**. By varying the burst lengths to match Morse code dot/dash patterns, information can be transmitted.

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🔵 **Morse Code Basics**

- **Dot (·)** — short burst
- **Dash (–)** — long burst (3× dot duration)
- **Morse K** = – · – (dash, dot, dash)

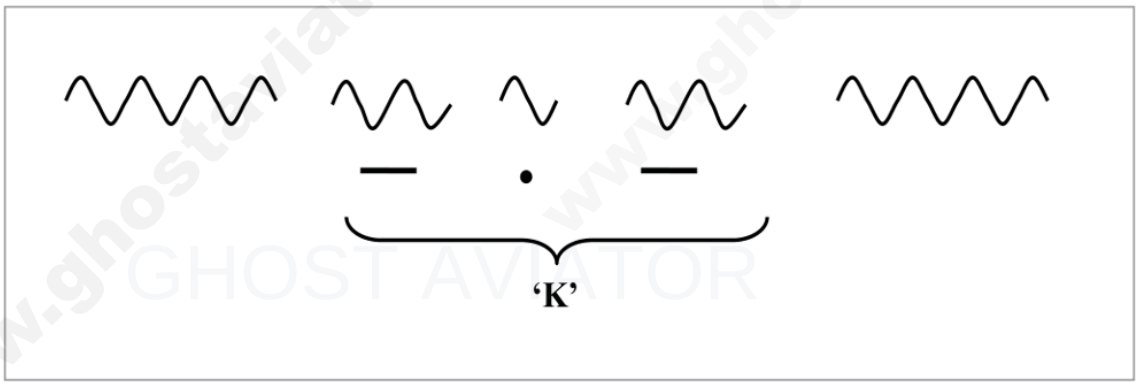


Figure 3.1 Morse 'K' in keyed modulation

Figure 3.1 — Morse 'K' in keyed modulation: the carrier wave is present for short (dot) and long (dash) periods in the pattern dash-dot-dash (K). Between transmissions, the carrier is silent

This form of transmission is called **telegraphy**. Until the development of AM, it was the only means of passing information by radio.

💡 **Aviation Relevance — NDB Identification**

Keyed modulation is still used today by **Non-Directional Beacons (NDBs)** for station identification. The NDB transmits its two or three-letter ICAO identifier in Morse code superimposed on the carrier. Pilots tune the ADF receiver, and the ident is heard in the headset, confirming the correct NDB is being received. This is covered in Chapter 7.

3. Amplitude Modulation (AM)

What this section covers: How the AF amplitude modifies RF amplitude, the heterodyning process, how sidebands are produced, bandwidth calculation, and power distribution in an AM signal.

In **Amplitude Modulation**, the amplitude of the audio frequency (AF) modifies (varies) the amplitude of the radio frequency (RF) carrier wave.

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AM — What Varies

AF amplitude
→ **RF amplitude**

FM — What Varies

AF amplitude
→ **RF frequency**

PM — What Varies

φ

AF amplitude
→ **RF phase**

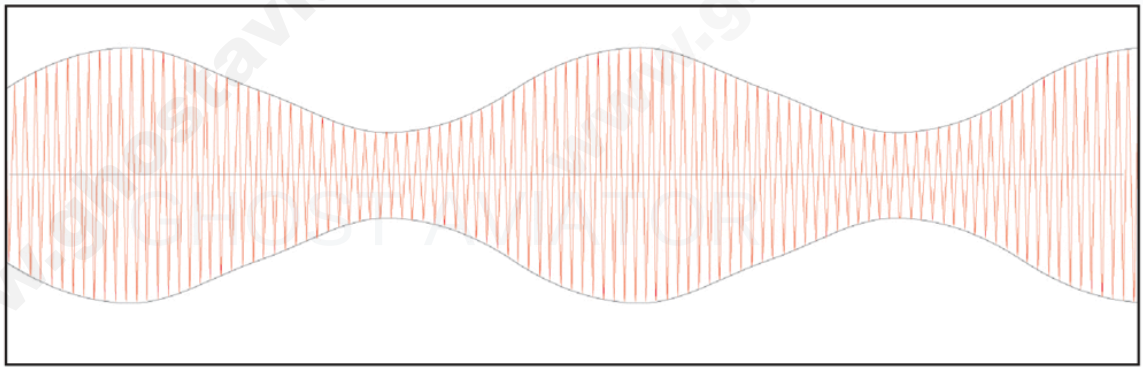


Figure 3.2 Amplitude modulation

Figure 3.2 — Amplitude modulation: the audio frequency (AF) signal varies the amplitude envelope of the RF carrier. Positive AF → RF amplitude increases; negative AF → RF amplitude decreases. The carrier frequency remains unchanged

3.1 Heterodyning & Sidebands

The process of combining RF and AF signals is called **heterodyning**. The result:

- The RF carrier frequency remains **unchanged**
- New frequencies are created at the **sum (RF + AF)** and **difference (RF − AF)** of the two frequencies
- These new frequencies form an **Upper Sideband (USB)** and a **Lower Sideband (LSB)**

📐 Worked Example — AM Sideband Calculation

```
RF = 2182 kHz  |  AF = 3 kHz
USB extends from: 2182.001 kHz → 2185 kHz   (RF + AF)
LSB extends from: 2181.999 kHz → 2179 kHz   (RF − AF)
Total spread: 2179 kHz → 2185 kHz
Bandwidth = 2185 − 2179 = 6 kHz = 2 × AF
```

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					2185 kHz	
				(25 W)	⇧	Upper Sideband (USB)
(100 W)	RF	2182 kHz			2182.001 kHz	
			⇨	(100 W)	2182 kHz	
(50 W)	AF	3 kHz			2181.999 kHz	
				(25 W)	⇩	Lower Sideband (LSB)
					2179 kHz	

Figure 3.3 AM sideband production

Figure 3.3 — AM sideband production: RF carrier at 2182 kHz (100W); AF of 3 kHz (50W) produces USB (25W) and LSB (25W). Total transmitted power = 150W. The carrier accounts for 2/3 of total power; information sidebands = 1/3 combined

AM BANDWIDTH FORMULA

```
Bandwidth = 2 × AF (audio frequency)

USB frequency = RF + AF

LSB frequency = RF − AF
```

📐 Exam-style AM Bandwidth Calculation

Q1: RF = 4716 kHz, AF = 6 kHz → Bandwidth = 2 × 6 = **12 kHz**

USB = 4716 + 6 = 4722 kHz; LSB = 4716 − 6 = 4710 kHz

Spread: 4710 kHz → 4722 kHz = 12 kHz ✓

3.2 AM Power Distribution

In a standard double-sideband AM transmission:

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Component	Power	% of Total	Contains Info?
RF Carrier	**100 W**	67%	No
Upper Sideband (USB)	**25 W**	17%	Yes (full copy)
Lower Sideband (LSB)	**25 W**	17%	Yes (full copy)
Total	**150 W**	100%	—

⚠ **Exam-Critical Efficiency Facts**

- Only **1/3 of total power** carries information (the two sidebands combined)
- The two sidebands contain **identical copies** of the audio information
- The carrier has **no information content** — it has done its job of shifting AF into RF frequencies
- Each sideband = **1/4 of carrier power** (25W vs 100W carrier)

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4. Single Sideband (SSB)

What this section covers: Why SSB is more efficient, how it works, and its three key advantages in HF/MF aviation communications.

Since both sidebands contain identical information, and the carrier contains no information, there is significant **redundancy** in standard double-sideband (DSB) AM. SSB solves this by:

1. Removing one sideband (either USB or LSB)
2. Suppressing the carrier
3. Transmitting only the remaining sideband — which contains **all** the information

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					2185 kHz	
				~~(25 W)~~ (150 W)	⇧	Upper Sideband (USB)
(100 W)	RF	2182 kHz			2182.001 kHz	
			⇨	~~(100 W)~~	~~2182 kHz~~	
(50 W)	AF	3 kHz			~~2181.999 kHz~~	
				~~(25 W)~~	⇩	~~Lower Sideband (LSB)~~
					~~2179 kHz~~	

Figure 3.4 Single sideband

Figure 3.4 — Single sideband (SSB): carrier and one sideband suppressed. The remaining sideband carries all 25W of useful information vs the DSB total of 150W. Power efficiency advantage is clearly visible

✅ **Three Advantages of SSB over DSB (Must Know All Three)**

1. **Double the channels:** Each SSB transmission uses only one sideband bandwidth → twice as many channels available in the same frequency band
2. **Better signal-to-noise ratio:** The ionosphere introduces static interference at MF/HF; narrower bandwidth of SSB significantly reduces interference from this static
3. **Less power required:** No power wasted on carrier or redundant sideband → lighter, more efficient equipment

Additionally, when using **sky wave propagation** (HF communications), different frequencies within the AM bandwidth refract by different amounts, causing distortion. The narrower bandwidth of SSB significantly reduces this distortion.

💡 **Why SSB for HF?**

HF frequencies (2–30 MHz) are in very high demand globally. SSB doubles the channel count, reduces power consumption, reduces static interference from the ionosphere, and reduces multipath distortion — all in one step. This is why **all HF radio telephony in aviation uses SSB** (emission designator J3E).

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5. Frequency Modulation (FM)

> **What this section covers:** How FM differs from AM, the relationship between AF amplitude/frequency and RF frequency deviation/rate of change, FM bandwidth figures, and why NBFM is not used in aviation.

In **Frequency Modulation**, the amplitude of the audio frequency modifies the **frequency** of the carrier wave (the amplitude of the carrier remains constant).

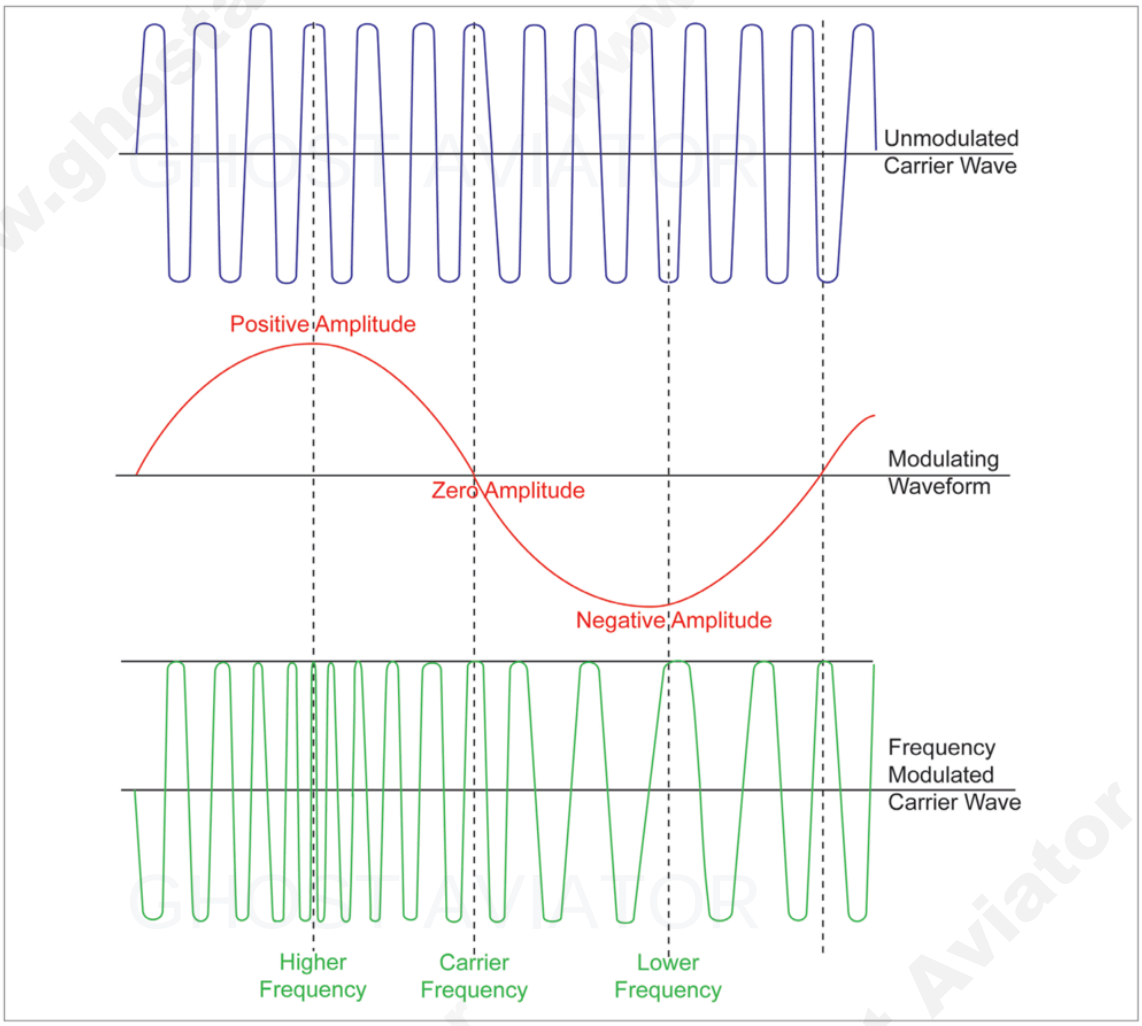


Figure 3.5 Frequency modulation

Figure 3.5 — Frequency modulation: the carrier wave frequency deviates above and below the nominal RF in proportion to the AF amplitude. Note that the amplitude of the modulated RF carrier remains constant throughout — unlike AM

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🔵 **Two FM Relationships — Both Examinable**

- **Frequency deviation** of carrier ∝ **amplitude** of AF (greater audio amplitude → greater frequency swing)
- **Rate of frequency change** = **frequency** of the AF modulating wave (higher audio pitch → faster frequency switching)

FM vs AM Bandwidth Comparison

System	Bandwidth	Notes
AM (DSB)	9 kHz	International standard allocation for AM broadcasting
FM Broadcast	150 kHz	By international agreement for music/broadcast
NBFM (voice)	8 kHz	Narrow Band FM — reduced bandwidth for voice only
Aeronautical Comms limit	6 kHz	ICAO standard for aeronautical voice channels
HF Comms limit	3 kHz	SSB occupies only one sideband

⚠️ **Why NBFM is NOT Used in Aviation**

Even NBFM (Narrow Band FM) with a reduced bandwidth of 8 kHz is still greater than the 6 kHz permitted for aeronautical voice communications. Therefore **NBFM communication systems are not used in aviation**. Standard aviation VHF/UHF uses AM (A3E for VHF, J3E for HF).

💡 **FM vs AM Summary**

- FM has **better noise immunity** than AM (noise appears as amplitude variations, which FM ignores)
- FM requires **much wider bandwidth** → impractical below VHF for aviation use
- FM is used in aviation navigation systems (e.g. VOR uses FM on the variable phase signal) but **not for voice comms**

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6. Phase Modulation

> **What this section covers:** Analogue phase modulation, digital phase shift keying (PSK), and the specific systems that use phase modulation in aviation.

In **Phase Modulation**, the **phase** of the carrier wave is modified by the input signal. There are two cases:

Analogue Phase Modulation

The phase of the carrier is modified in proportion to the **amplitude** of the analogue input signal — similar to AM but it is the phase angle that changes rather than the amplitude or frequency.

Digital Phase Shift Keying (PSK)

With a digital (binary) input signal, the phase change reflects a binary **0 or 1**:

> 🔵 **Binary PSK (simplest case)**
>
> - **0° phase shift** → represents binary **0**
> - **180° phase shift** → represents binary **1**
> - More complex systems use multiple phase shift values to represent more data per symbol (e.g. QPSK = 4 phase states, 8PSK = 8 phase states)

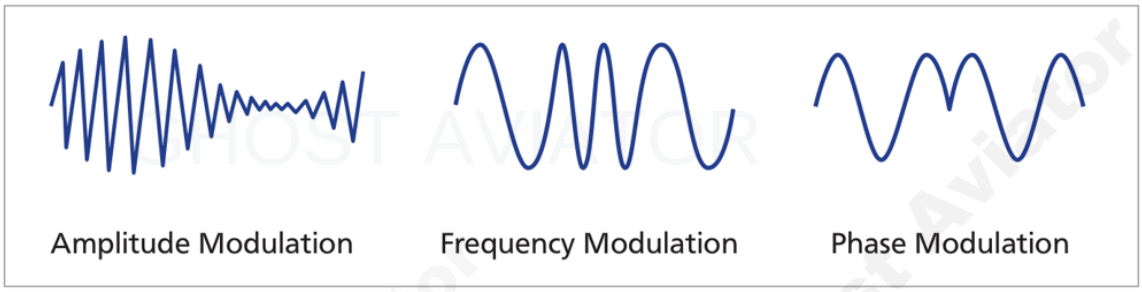


Figure 3.6

Figure 3.6 — Side-by-side comparison of Frequency Modulation (left), Phase Modulation (centre), and Amplitude Modulation (right). In FM: frequency varies, amplitude constant. In PM: phase shifts discretely, amplitude constant. In AM: amplitude varies, frequency constant

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✅ **Aviation Systems Using Phase Modulation**

- **GPS** — uses **Binary Phase Shift Keying (BPSK)** to transmit navigation data and ranging codes
- **MLS (Microwave Landing System)** — uses **Differential Phase Shift Keying (DPSK)**

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7. Pulse Modulation

What this section covers: How pulse modulation works and where it is used in aviation.

Pulse modulation is formed by generating and transmitting a sequence of short-period pulses — the carrier wave is transmitted in intermittent bursts rather than continuously.

🔵 **How Pulse Modulation Works**

- Carrier wave is transmitted as a series of **brief, high-power pulses**
- Between pulses, the transmitter is silent — it can listen for replies
- Information can be encoded in the **timing, amplitude, width, or position** of pulses

✅ **Aviation Systems Using Pulse Modulation**

- **Radar systems** (primary radar, SSR, AWR) — pulse transmitted, reflected pulse timed to determine range
- **DME (Distance Measuring Equipment)** — pulse pairs transmitted/received to measure slant range (emission designator P0N)
- **SSR (Secondary Surveillance Radar)** — pulse trains for interrogation and reply

8. Emission Designators

What this section covers: How to decode three-character emission designators, and the specific designators for key aviation equipment.

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Emission designators are a standardised three-character code (letter-digit-letter) that describe the characteristics of an electronic transmission. You are not required to memorise the full table, but you must know the key aviation examples.

EMISSION DESIGNATOR STRUCTURE

```
Symbol 1 (Letter): Type of modulation of main carrier
Symbol 2 (Digit): Nature of signals used for modulation
Symbol 3 (Letter): Type of information transmitted
```

Symbol 1 — Modulation Type

Letter	Modulation
N	Unmodulated carrier (no modulation)
A	Amplitude Modulation — Double Sideband (DSB)
H	AM — Single Sideband, Full Carrier
J	AM — Single Sideband, Suppressed Carrier (SSB)
F	Frequency Modulation
G	Phase Modulation
P	Sequence of unmodulated pulses
K	Sequence of pulses modulated in amplitude

Symbol 2 — Nature of Modulating Signal

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Digit	Signal Type
0	No modulating signal
1	Single channel — quantized/digital information, no sub-carrier
2	Single channel — quantized/digital information, with sub-carrier
3	Single channel — analogue information
7	Two or more channels — quantized/digital information
8	Two or more channels — analogue information
9	Composite: digital + analogue channels

Symbol 3 — Information Type

Letter	Information Transmitted
N	No information
A	Telegraphy for aural reception (Morse)
B	Telegraphy for automatic reception
D	Data transmission, telemetry, telecommand
E	Telephony (including sound broadcasting)
W	Combination of above
X	Cases not otherwise covered

Key Aviation Emission Designators — Worked Examples

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📐 A3E — VHF Radio Telephony

A = Amplitude Modulation, Double Sideband

3 = Single channel, analogue information

E = Telephony

→ AM carrier wave modulated with speech (VHF voice comms)

📐 J3E — HF Radio Telephony

J = AM, Single Sideband, Suppressed Carrier (SSB)

3 = Single channel, analogue information

E = Telephony

→ SSB voice comms over HF (carrier + one sideband removed)

Equipment / System	Designator	Meaning
ADF / NDB	`N0NA1A / N0NA2A`	Unmod carrier + keyed Morse ident
VHF RTF	`A3E`	AM-DSB, analogue, telephony
HF RTF	`J3E`	SSB suppressed carrier, telephony
VOR	`A9W`	AM, composite analogue+digital, combination
ILS	`A8W`	AM, 2+ analogue channels, combination
Marker Beacons	`A2A`	AM, digital sub-carrier, Morse ident
DME	`P0N`	Unmod pulse sequence, no info (ranging only)
MLS	`N0XG1D`	Phase mod, digital data, guidance

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⚠ EXAM NOTE

"It is not necessary to know the details of the [full emission] table." However you **must** know **A3E** (VHF RTF) and **J3E** (HF RTF) and be able to decode them. Other designators (ADF, VOR, ILS, DME) are "unlikely to be examined" but may appear as distractors.

⚡ QUICK REVISION SUMMARY — CHAPTER 3

- **Keyed:** interrupts carrier; Morse code; used by NDBs for ident
- **AM:** AF amplitude → RF amplitude; bandwidth = 2 × AF; USB = RF+AF, LSB = RF−AF
- **AM power:** carrier = 2/3 total; each sideband = 1/6 total; only 1/3 carries info
- **SSB advantages:** 2× channels, better S/N, less power
- **FM:** AF amplitude → RF frequency deviation; AF frequency → rate of change
- **FM bandwidth:** Broadcast = 150 kHz; NBFM = 8 kHz > 6 kHz limit → NOT used in aviation voice
- **Phase modulation:** analogue (amplitude → phase angle); digital PSK (0° = 0, 180° = 1)
- **GPS = BPSK; MLS = DPSK**
- **Pulse modulation:** radar systems (DME, SSR, primary radar)
- **A3E** = VHF RTF; **J3E** = HF RTF (SSB suppressed carrier)
- **DME = P0N** (pulse, unmod, no info — pure ranging)

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📝 Practice Questions & Detailed Answers

How to Use This Section

Cover the answer panel and attempt each question independently. Read the full explanation and distractor analysis — knowing *why* the wrong answers are wrong is as important as knowing the right answer.

Q1. The bandwidth produced when a radio frequency (RF) of 4716 kHz is amplitude modulated with an audio frequency (AF) of 6 kHz is:

a. 6 kHz

b. 3 kHz

c. 12 kHz

d. 9 kHz

✓ Correct Answer: **(c)** — 12 kHz

Explanation: AM bandwidth = 2 × AF = 2 × 6 kHz = **12 kHz**. The USB extends from 4716 to 4722 kHz; the LSB extends from 4716 to 4710 kHz. Total spread = 4710–4722 kHz = 12 kHz. The RF carrier frequency (4716 kHz) is irrelevant to the bandwidth calculation. See Section 3.1.

Why the other options are wrong:

- **(a)** — 6 kHz = the AF itself. This confuses the modulating frequency with the resulting bandwidth. Bandwidth = 2× AF, not 1×.
- **(b)** — 3 kHz = half the AF. No basis for this value in AM calculations.
- **(d)** — 9 kHz = standard AM broadcasting allocation, but it is not the answer to this specific calculation with AF = 6 kHz.

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Instructor's Note: *The classic trap is answering 6 kHz (the AF itself). Always: Bandwidth = 2 × AF. The RF value is only needed if asked for the USB or LSB frequency. This is the most likely calculation to appear in the DGCA exam for this chapter.*

Q2. Which of the following statements concerning AM is correct?

a. The amplitude of the RF is modified by the frequency of the AF

b. The amplitude of the RF is modified by the amplitude of the AF

c. The frequency of the RF is modified by the frequency of the AF

d. The frequency of the RF is modified by the amplitude of the AF

✓ Correct Answer: **(b)** — The amplitude of the RF is modified by the amplitude of the AF

Explanation: In AM, the **amplitude of the AF** modifies the **amplitude of the RF**. When AF amplitude is high → RF amplitude is high; when AF amplitude is low → RF amplitude is low. The RF frequency remains unchanged. See Section 3.

Why the other options are wrong:

- **(a)** — "frequency of AF modifies amplitude of RF" — this is nonsense for AM. AF frequency determines the frequency of the sidebands, not the RF amplitude.
- **(c)** — "frequency of RF modified by frequency of AF" — this describes neither AM nor FM.
- **(d)** — "frequency of RF modified by amplitude of AF" — this is the definition of **FM**, not AM. A very common exam trap.

Instructor's Note: *Options (b) and (d) are the key pair here. (b) = AM definition; (d) = FM definition. These two are systematically swapped in wrong-answer*

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options. Memorize: AM → amplitude changes amplitude; FM → amplitude changes frequency.

Q3. Which of the following is an advantage of single sideband (SSB) emissions?

a. More frequencies available

b. Reduced power requirement

c. Better signal/noise ratio

d. All of the above

✓ Correct Answer: **(d)** — All of the above

Explanation: SSB has **three** advantages: (1) more frequencies/channels (each SSB uses half the bandwidth of DSB); (2) reduced power requirement (no power wasted on carrier or second sideband); (3) better signal/noise ratio (narrower bandwidth, less ionospheric static). All three options (a), (b), and (c) are correct individually, so (d) is the best answer. See Section 4.

Why the other options are not the best:

- **(a)** — True but incomplete. More channels is one advantage, but not the only one.
- **(b)** — True but incomplete. Reduced power is one advantage.
- **(c)** — True but incomplete. Better S/N ratio is one advantage.

Instructor's Note: *Any time a question asks "which is an advantage" and lists multiple valid advantages individually plus "all of the above" — the answer is almost certainly "all of the above." Know all three SSB advantages cold: channels × 2, power ↓, S/N ↑.*

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Q4. Which of the following statements concerning FM is correct?

a. The amplitude of the RF is modified by the frequency of the AF

b. The amplitude of the RF is modified by the amplitude of the AF

c. The frequency of the RF is modified by the frequency of the AF

d. The frequency of the RF is modified by the amplitude of the AF

✓ Correct Answer: **(d)** — The frequency of the RF is modified by the amplitude of the AF

Explanation: In FM, the **amplitude of the AF** determines the **frequency deviation** of the RF carrier. A large AF amplitude → large frequency swing; small AF amplitude → small frequency swing. The RF amplitude remains constant. See Section 5.

Why the other options are wrong:

- **(a)** — "amplitude of RF modified by frequency of AF" — neither AM nor FM description. This has no basis.
- **(b)** — "amplitude of RF modified by amplitude of AF" — this is the definition of **AM**, not FM. The paired trap with Q2.
- **(c)** — "frequency of RF modified by frequency of AF" — partially relates to FM (AF frequency = rate of change of carrier frequency deviation), but this is NOT the primary FM relationship. The frequency of the modulating signal determines the RATE of carrier frequency change, not the magnitude of frequency deviation. Subtle but wrong.

Instructor's Note: *Option (c) is the subtle trap. AF frequency does affect FM — it determines the rate of change — but the defining relationship for FM is that AF amplitude determines the magnitude of the frequency deviation. The question asks what "modifies" the RF frequency, and the answer is AF amplitude (not AF frequency). Option (d) is unambiguously correct.*

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📊 Master Reference Tables

Bandwidth Summary

System / Standard	Bandwidth	Notes
AM (DSB) formula	**2 × AF**	Upper + lower sideband combined
AM Broadcasting (standard)	**9 kHz**	International allocation
FM Broadcasting	**150 kHz**	International agreement for music radio
Narrow Band FM (NBFM)	**8 kHz**	Voice only, NOT used in aviation
Aeronautical Comms limit	**6 kHz**	ICAO standard channel spacing limit
HF Comms (SSB)	**3 kHz**	Single sideband only

Modulation Comparison

Type	What AF Does to Carrier	What AF Amplitude Controls	Aviation Use
Keyed	Switches carrier on/off	N/A	NDB ident
AM (DSB)	Varies RF amplitude	RF amplitude	VHF RTF (A3E)
SSB	Varies RF amplitude (1 SB only)	RF amplitude	HF RTF (J3E)
FM	Varies RF frequency	Frequency deviation (swing)	Not for voice; VOR signal phase
Phase (PSK)	Varies RF phase	Phase angle shift	GPS (BPSK), MLS (DPSK)
Pulse	Switches carrier on/off (brief pulses)	Pulse timing / width	DME, Radar, SSR

Answer Key — Chapter 3 Questions

Q1	Q2	Q3	Q4

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c	b	d	d

Mnemonics & Memory Aids

- **AM:** "Amplitude Modifies Amplitude" — AF amplitude → RF amplitude
- **FM:** "Frequency has Fluctuating frequency" — AF amplitude → RF frequency deviation
- **Bandwidth = 2 × AF** — always double the audio frequency
- **AM power:** "Carrier = 2/3, sidebands = 1/3 total; but info is only in sidebands"
- **SSB triple advantage:** "2× Channels, Better S/N, Less Power" → C-S-P
- **NBFM NOT in aviation:** 8 kHz > 6 kHz limit
- **GPS = BPSK; MLS = DPSK** — both Phase Shift Keying
- **A3E** = VHF voice; **J3E** = HF voice (SSB); **P0N** = DME pulse

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